

Electrochemical Synthesis of Nanoporous Platinum Nanoparticles Using Laser Pulse Heating: Application to Methanol Oxidation

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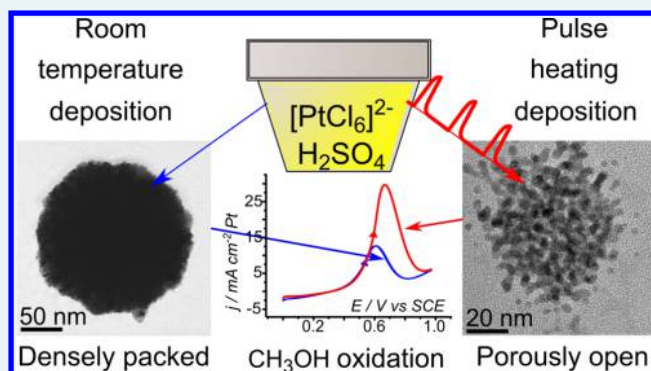
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S Supporting Information

ABSTRACT: Nanoporous platinum nanoparticles (NPs) have been proposed as promising electrocatalytic materials. Routes to produce them typically consist of chemical synthesis or selective dissolution of one component of a two-component mix. Here we show that by employing a pulsed laser heating approach during electrodeposition, whereby the electrode/electrolyte interface is continually heated and cooled, NPs with a nanoporous structure can be grown directly on the electrode (boron-doped diamond) surface. Transmission electron microscopy shows the NPs to be composed of loosely packed aggregates of much smaller crystalline particles of size 2–5 nm, with the porosity increasing with increasing deposition overpotential. In contrast, electrodeposition at room temperature (RT) results in particles which show a considerably more compact morphology and fewer higher index crystal facets, as revealed by electron diffraction techniques. Pulsed heating also offers a route toward controlling the monodispersity of the electrodeposited NPs. When applied to the oxidation of methanol, the laser-heated NPs show considerably higher catalytic current densities in comparison to RT-deposited particles. The highest catalytic activity is observed for the most porous NPs produced at the highest overpotential. Interestingly, the ratio of the forward oxidative current to the backward current is highest for those particles deposited under laser-heated conditions but with the smallest overpotential. This suggests that the most catalytically active NPs may also encourage binding of residual adsorbed carbon monoxide and that a compromise must be reached.

KEYWORDS: nanoporous, boron-doped diamond, Pt electrodeposition, electrocatalysis, electron microscopy, methanol oxidation



1. INTRODUCTION

Nanosized particles of platinum (Pt) supported on an electrode are pivotal for an array of electrocatalytic reactions, such as the oxidation–reduction reactions in direct methanol (MeOH) fuel cells, a subcategory of the proton exchange membrane fuel cell.^{1,2} The electrocatalytic activity of the Pt nanostructure is governed by the morphology, size, loading level, and stability of the deposit, as well as any possible interactions between the support substrate and particle which facilitate electron transfer.^{3,4}

Electrodeposition is a versatile method for both fabricating the nanostructure and ensuring it remains in intimate electrical contact with the underlying electrode substrate.^{5–7} The resulting morphology depends on the deposition parameters employed such as potential, time, and choice of substrate.⁸ Temperature may also have a significant effect on the structure,⁸ although surprisingly this is often left unexplored in the academic literature.

In practice, temperature can be controlled either isothermally or nonisothermally.⁹ The former often involves heating the solution using a water jacket or bath, but this approach can lead to uncontrolled mass transport, ill-defined thermal gradients,

and slow response to changes in temperature.⁹ Conversely, a nonisothermal heating approach is attractive due to significantly greater control over mass transport in the system, faster heating times, and the ability to heat beyond the boiling point of water without introducing unwanted bubbles or solvent evaporation.¹⁰

Nonisothermal heating has been achieved previously using a pulsed laser.^{11,12} Typically, the electrode surface is illuminated through solution,^{13,14} although this can result in unwanted photoeffects, such as electrolyte decomposition and formation of unknown compounds as side products, in addition to oxide formation.¹¹ There are only limited reports of laser heating via irradiation of the back surface of the electrode.^{15–17} Depending on the response of the electrode material to the input thermal energy,¹⁸ both approaches can result in unwanted damage to the electrode surface.

Over the past decade, boron-doped diamond (BDD) has emerged as an interesting choice as a substrate for electro-

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deposition due to its sp^3 -bonded structure and consequently inert surface,¹⁹ leading to low background currents and a wide solvent window, which are also attractive for electroanalytical applications.²⁰ BDD also has a very high thermal diffusivity ($600 \text{ W m}^{-1} \text{ K}^{-1}$ at 300 K) and extreme resistance to thermal ablation, being able to withstand laser power densities, P_d , of GW cm^{-2} (for nanosecond pulses)^{21–23} and is thus an ideal material for thermoelectrochemical studies.^{24,25} For BDD, absorption of optical energy from a laser occurs very quickly, resulting in rapid heating of the surface.^{24,25} Thus, the interface temperature can be changed elegantly without changing the temperature of the bulk solution. The BDD surface can also be either grown or prepared (by polishing) to nanometer-level roughness, allowing electrodeposited nanoparticles (NPs) to be probed with techniques such as atomic force microscopy (AFM) and field emission scanning electron microscopy (FE-SEM).²⁰

Commonly in electrochemical deposition the resulting particle size and shape are often heterogeneous due to differing rates of nucleation and growth.²⁶ This situation is further complicated with BDD, as the material is typically employed in the polycrystalline form, which is heterogeneously doped.²⁰ Attempts to electrodeposit relatively monodisperse metal NPs have been demonstrated (on highly oriented pyrolytic graphite, HOPG), using a double potential step (DPS) technique to separate growth and nucleation.²⁷ By combining DPS with tailored solution conditions such as pH and electrolyte chemical composition through e.g. the addition of capping agents and surfactants, it is also possible to control the shape of the electrodeposited particles.^{28,29}

Among the many shapes and structures possible for Pt NPs,³⁰ porous Pt has gained recent interest due to its high electrocatalytic activity,^{30,31} attributed primarily to the high surface area to volume ratio³¹ and the creation of a confined reaction environment.³² Methods to fabricate nanoporous Pt NPs have thus become an important topic of research. To date, porous Pt has been synthesized in solution, using transmetalation reactions, self-organization processes, and templating methods.^{33–36} To achieve such structures electrochemically, the most common method is electrodeposition of bimetallic NPs, followed by preferential leaching of the least noble metal.³⁷ Recently, DPS electrodeposition has also been used to both control the size range of the deposited particles and induce porosity.³¹

In this study, we explore whether electrodeposition under controlled, elevated-temperature conditions also offers a route toward the creation of nanoporous structures. BDD is used as the substrate electrode, and a pulsed laser approach is utilized to nonisothermally heat the electrode, from the back side, during electrodeposition of Pt. We also examine the subsequent electroactivity of the resulting structures, produced under both room temperature (RT) and laser pulse heating, toward the electro-oxidation of MeOH.

2. EXPERIMENTAL SECTION

2.1. Solutions and Chemicals. Pt electrodeposition solutions were prepared from potassium hexachloroplatinate (K_2PtCl_6 , 99%, Sigma-Aldrich) with sulfuric acid (H_2SO_4 , 99.99%, Sigma-Aldrich) as a supporting electrolyte in ultrapure Milli-Q water ($18.2 \text{ M}\Omega \text{ cm}$, Millipore Corp., U.S.) at 20°C . All chemicals were used as received without further purification. Pt deposition solutions contained $500 \mu\text{M} [\text{PtCl}_6]^{2-}$ in $0.5 \text{ M H}_2\text{SO}_4$ (pH 0.65) which was deaerated with N_2 gas for 15 min

before the experiment to exclude O_2 . N_2 flow was maintained over the solution during the experiment.

2.2. Materials. The experiments were performed using a polycrystalline BDD plate, 9 mm in diameter and $200 \mu\text{m}$ thick, with a boron dopant level of $\sim 3 \times 10^{20} \text{ B atoms cm}^{-3}$, grown using microwave chemical vapor deposition by Element Six (Harwell, Oxford, U.K.).²⁵ To make an ohmic contact to the electrode, the upper quarter of the back surface was sputtered (Moorfield MiniLab 060 Platform) with Ti (20 nm)/Au (400 nm) and annealed in a tube furnace for 5 h at 450°C .³⁸

2.3. Electrochemical Setup. For the electrochemical measurements during pulsed heating and at RT, a custom-built electrochemical cell was designed and fabricated in house. The cell is composed of two parts, a Teflon container and a Perspex window, held together with Teflon screws. The Perspex window contained a hole in the center, across which the BDD disk was mounted (inside face) and held in place with adhesive Kapton tape (100 μm thickness, stable up to 1000°C , R.S. Components Ltd.). The geometry of the electrode area exposed to solution was defined using Kapton tape, which contained a laser-micromachined circular hole that was 1 mm in diameter. A conductive track was made from the Au-sputtered contact on the back face of the BDD electrode (not in contact with solution) using two parts of silver conductive epoxy (R.S. Components Ltd.) to the top of the Perspex window, where electrical contact was made with a crocodile clip. The setup is shown in Figure 1.

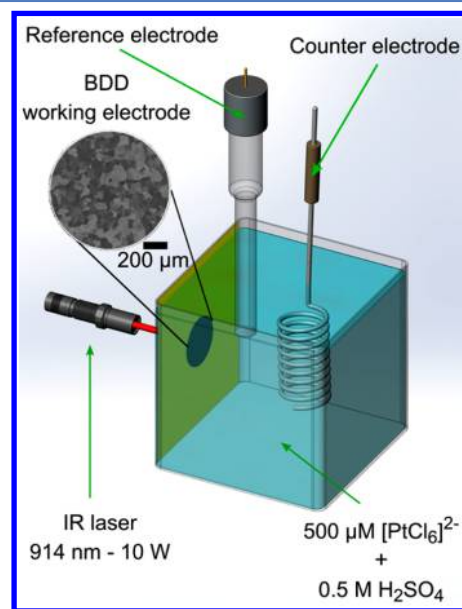


Figure 1. Schematic drawing of the pulsed laser heating experimental setup. The laser beam is focused by the laser lens onto the back face of the BDD electrode held in the Perspex cell with Kapton tape, which also defines the active electrode area (1 mm disk). The inset image is an FE-SEM image of the BDD electrode surface, where the different colors represent the differently doped grains.

For the nonisothermal experiments, cyclic voltammetry (CV) and chronoamperometry were performed simultaneously with pulsed laser heating, termed temperature pulse voltammetry (TPV). A diode IR laser (914.7 nm, 30 W, LM-D0296, LIMO) was used for all pulsed temperature experiments, controlled by a laser diode controller (LDC1000, Laser Electronics Ltd. Lincolnshire, U.K.). An optical fiber was used to deliver the

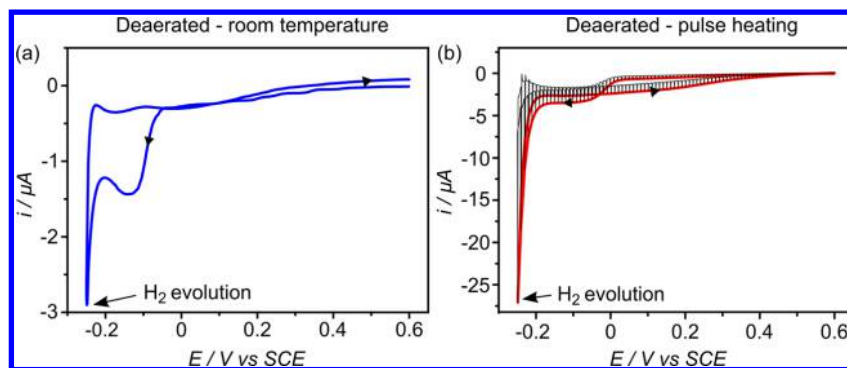


Figure 2. CVs for Pt electrodeposition recorded in a deaerated solution containing 500 μM $[\text{PtCl}_6]^{2-}$ in 0.5 M H_2SO_4 at 0.1 V s^{-1} under (a) RT and (b) laser-heated conditions (1.2 kW cm^{-2} , pulsed laser on for 10 ms and off for 90 ms).

irradiation of a 1 mm diameter central circular spot via a focusing lens onto the back face of the BDD electrode. A dual-channel function generator (AFG3022B, Tektronix) was employed as a master controller for TPV experiments to ensure that the potentiostat (CompactStat, Ivium Technologies, The Netherlands) and laser were synchronized when triggered at the start of the experiment. For these experiments, a fixed laser power (P_d) of 1.2 kW cm^{-2} was utilized (laser power 10 W) and the laser was cycled with 10 ms on and 90 ms off, i.e. a heating period of 10 ms and a cooling period for 90 ms, to ensure that heat does not dissipate to the bulk solution. This P_d value equated to an initial average local temperature at the electrode/electrolyte interface of 66.4°C , determined via open circuit potential (OCP) measurements in a solution containing 0.5 mM potassium ferricyanide, $\text{Fe}(\text{CN})_6^{3-}$ (Sigma-Aldrich, 99%), and 0.5 mM potassium hexacyanoferrate trihydrate, $\text{Fe}(\text{CN})_6^{4-}$ (Sigma-Aldrich, $\geq 98.5\%$), the method for which has been described previously in detail.^{24,25}

All electrochemical experiments were carried out using a three-electrode setup controlled by a potentiostat: CompactStat Ivium Technologies (The Netherlands) for TPV and CHI730A (CH Instruments, Inc. Austin, TX) for RT experiments. A commercial saturated calomel electrode (SCE; CHI150, CH Instruments Inc., Austin, TX) was used as the reference electrode and a helical Pt wire served as the counter electrode. For hydrogen adsorption/desorption studies in sulfuric acid, mercurous sulfate, MSE ($\text{Hg}/\text{HgSO}_4/\text{K}_2\text{SO}_4(\text{sat.})$), served as the reference electrode, to avoid any possible Cl^- contamination/adsorption affects.^{39,40}

For all experiments, to ensure a clean BDD surface, the electrode was first polished with alumina powder ($0.05 \mu\text{m}$ particles, micropolish, Buehler, Germany) and then further polished on a water-saturated polishing pad (micropolish, Buehler, Germany). The electrode was then cycled in 0.5 M H_2SO_4 for 5 min to verify a clean surface CV response.

2.4. Surface Characterization. The morphology of the electrodeposited Pt NPs was characterized ex situ using AFM, FE-SEM, and transmission electron microscopy (TEM). Before imaging, the surface of the sample was rinsed with deoxygenated ultrapure water and left to dry in a desiccator under a N_2 atmosphere. For each experiment at least three images ($n = 3$) were recorded in different areas of the surface. AFM images were acquired at a low scan rate (0.25 Hz) in tapping mode (TM) using a Bruker Enviroscope AFM with a Nanoscope IV controller. FE-SEM images were recorded using the secondary electron (SE2) detector on a Zeiss Supra 55-VP instrument operating at 7 keV. Energy dispersive X-ray

spectroscopy (EDXS) spectra were recorded using an Oxford Instruments Si-Li detector unit on the FE-SEM instrument, at a working distance of 8.5 mm and accelerating voltage of 10 keV. TEM measurements of particle shape, structure, size, and composition were carried out using a JEOL-JEM 2100FX (LaB₆ filament) microscope operated at 200 keV.

For TEM imaging, a positive replica technique was employed to transfer the Pt NPs to a Cu TEM grid (300 mesh, Agar Scientific).⁴¹ This procedure is shown schematically in Figure S1 of the Supporting Information and consists of the following steps. (i) A drop of acetone was placed on the Pt NP electrodeposited BDD surface, followed immediately by the replica film (cellulose acetate sheet), pulled onto the surface by surface tension. The film was left to dry for 30 min. (ii) The film was separated from the surface and placed on a microscope slide with the NPs facing upward. (iii) The surface was then vacuum carbon coated (Emitech evaporator, Quorum Technologies, U.K.). (iv) The required area was cut from the film and laid, carbon side up, on the TEM grid. (v) The grid was subsequently placed in acetone to dissolve the cellulose sheet, leaving the NPs supported on the carbon film attached to the Cu TEM grid. (vi) After 1 h, the grid was removed from the mesh and dried in a desiccator under N_2 flow before being placed in the TEM for examination. In addition to imaging, EDXS was also carried out to determine the composition of the deposited material. In order to minimize electron beam damage, all images were acquired with one single scan fast exposure with a beam current of $\sim 10 \text{ pA}$ and an acquisition dwell time of $12 \mu\text{s}$ per pixel.

3. RESULTS AND DISCUSSION

3.1. CV Studies of Pt Electrodeposition in Acid Solution at Room and Elevated Temperature. The speciation of Pt when $\text{Pt}(\text{IV})$ salts are dissolved in aqueous media can be influential in the electrodeposition process.⁴² A low pH was chosen to make hydrated $\text{Pt}(\text{IV})$ the major solution species.⁴³ In order to understand the characteristic electrochemical signature of $[\text{PtCl}_6]^{2-}$ in 0.5 M H_2SO_4 (at pH 0.65) and the role of an increased interfacial temperature on $\text{Pt}(\text{IV})$ reduction, CVs were carried out at RT ($T = 20.0^\circ\text{C}$) and under laser pulse heating (average T across the electrode/electrolyte interface 66.4°C), under deaerated conditions, as shown in Figure 2. The CVs were run negatively at 0.1 V s^{-1} from +0.6 V (where no Faradaic electron transfer reactions take place) to -0.25 V (where hydrogen evolution begins to dominates) and back to +0.6 V.

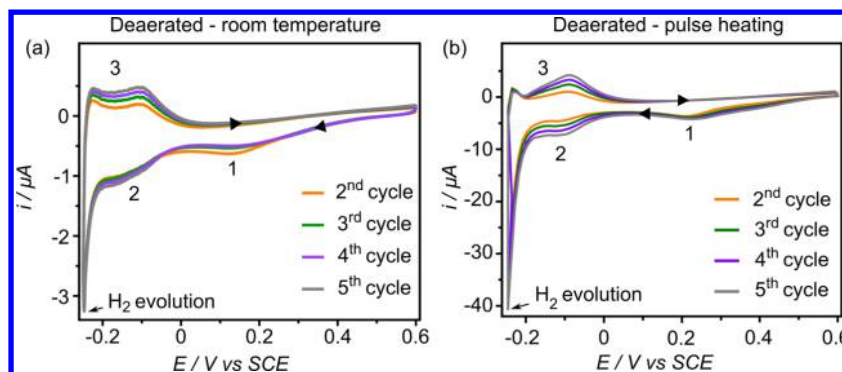


Figure 3. CVs of four subsequent cycles (2nd to 5th) following the first cycle of $[\text{PtCl}_6]^{2-}$ electrochemistry (shown in Figure 2) under (a) RT and (b) heating by laser pulsing in a deaerated solution containing $500 \mu\text{M}$ $[\text{PtCl}_6]^{2-}$ in $0.5 \text{ M H}_2\text{SO}_4$ at 0.1 V s^{-1} . Note that the CVs recorded under laser pulse heating represent the maximum current achieved during each current pulse.

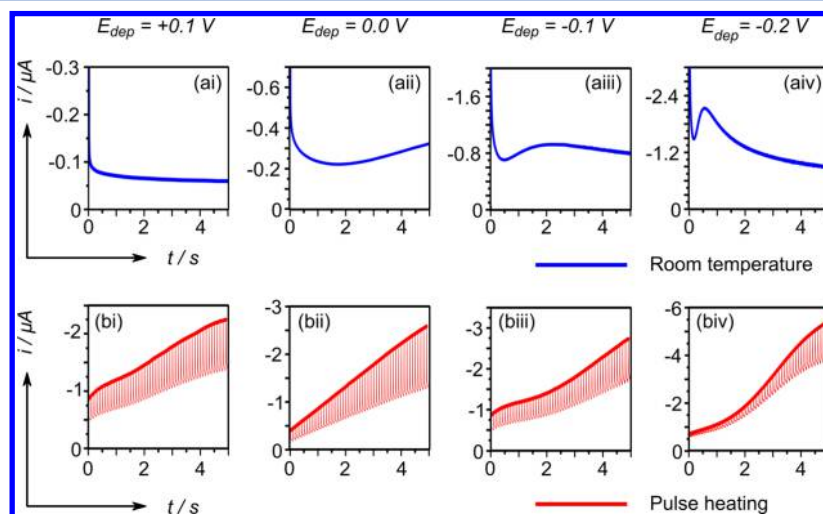


Figure 4. Current–time transients for Pt electrodeposition at (a) RT (blue lines) and (b) laser pulse heating (red lines) at E_{dep} values of (i) $+0.1 \text{ V}$, (ii) 0 V , (iii) -0.1 V , and (iv) -0.2 V for $t_{\text{dep}} = 5 \text{ s}$. The bold red line represents the maximum current passed during the laser pulse.

Figure 2a shows the response under RT conditions (blue line). In Figure 2b, under laser-heated conditions, the pulsed response of the laser (10 ms on, 90 ms off) generates a pulse-like current response (black line). During the heating period a temperature gradient extends out from the electrode surface into the solution nearby, which then readjusts during the cooling period. The red line in Figure 2b represents the maximum current achieved during each laser pulse.

Under RT conditions (Figure 2a), scanning negatively (forward scan) leads to an increase in current from ca. $+0.5 \text{ V}$ vs SCE, due, most likely, to the onset of Pt reduction on the BDD (kinetically limited region). As the potential is made more negative, and Pt reduction moves into a diffusion-limited region, a current peak of $-1.48 \mu\text{A}$ is seen at -0.11 V . At potentials more negative than -0.21 V a large increase in current is seen due to the reduction of protons (H^+) and subsequent evolution of hydrogen (H_2). When the scan direction is reversed, the current still flows in the negative direction due to further reduction of Pt on the freshly formed Pt phase.⁴⁴

Under laser-heated conditions (Figure 2b), scanning negatively produces features similar to those observed under RT, except the currents passed are larger. This may be due to a combination of many factors, including (i) the increased electron transfer kinetics for reduction of Pt(IV) and H^+ ,¹¹ (ii) an increase in diffusional flux due to both a reduction in local

viscosity and increase in diffusion coefficient (in accordance with the Stokes–Einstein equation¹¹) in the vicinity of the electrode/electrolyte interface, and (iii) convective effects arising from density changes due to the established thermal gradient.^{24,25,45} The potential at which a current starts to flow in the diffusion-limited regime also occurs at a slightly more positive potential in comparison with RT conditions (-0.07 V versus -0.11 V , Figure 2), suggesting increased ET kinetics and/or a negative temperature coefficient, for Pt deposition under high-temperature conditions.^{24,25} When the scan direction is reversed, a negative current is again seen in the pulsed CV.

The results of repeated scanning over a further four cycles, where Figure 2 represents cycle 1, at 0.1 V s^{-1} at RT and with laser pulse heating in the acidic Pt electrodeposition solution are shown in parts a and b of Figure 3, respectively. As Pt is continuously deposited on the surface during the additional four CV cycles (peak 1),^{46–48} the expected CV response associated with the presence of polycrystalline Pt in acidic solution (peaks 2 and 3) emerges.^{49,50} In particular, Figure 3a shows clearly the H adsorption (H_{ads}) and H desorption (H_{des}) peaks, in the region between -0.01 and -0.21 V . H_{ads} (peak 2) is associated with a negative current ($\text{H}^+ + \text{e}^- \rightarrow \text{H}_{\text{ads}}$), while H_{des} (peak 3) is associated with a positive current ($\text{H}_{\text{des}} \rightarrow \text{H}^+ + \text{e}^-$).⁵¹ As the potential is scanned positively, the Pt will oxidize to form PtO_x provided a sufficient positive potential is reached;

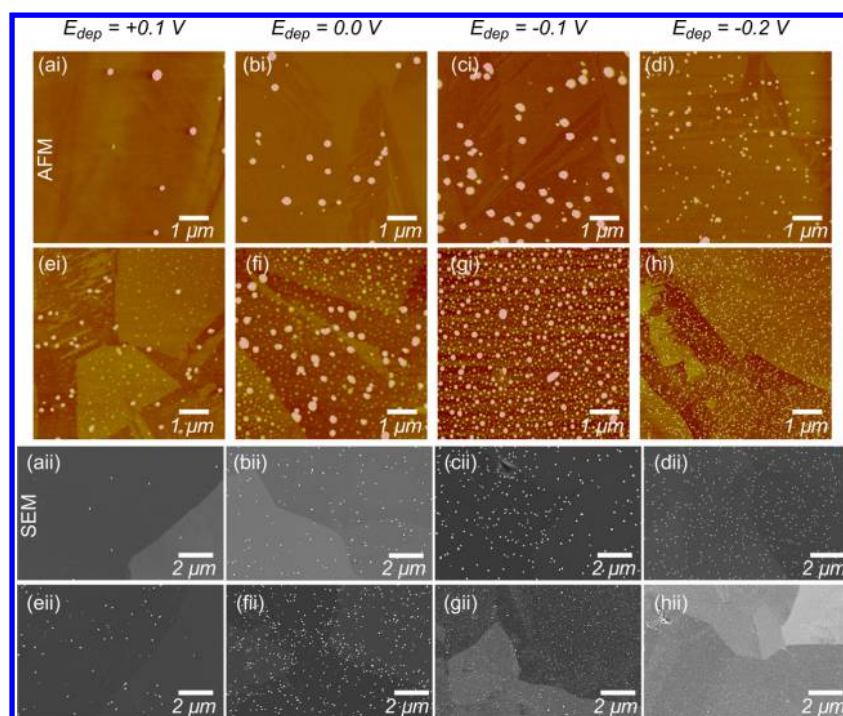


Figure 5. Representative (i) $5\ \mu\text{m} \times 5\ \mu\text{m}$ AFM and (ii) FE-SEM images of a BDD electrode after electrodeposition of Pt NPs from a solution containing $500\ \mu\text{M}\ \text{PtCl}_6^{2-}$ and $0.5\ \text{M}\ \text{H}_2\text{SO}_4$ (a–d) at RT and (e–h) under laser pulsed heating. Electrodeposition was carried out by applying $E_{\text{dep}} = +0.1\ \text{V}$ (a and e), $0\ \text{V}$ (b and f), $-0.1\ \text{V}$ (c and g), and $-0.2\ \text{V}$ (d and h) for $t_{\text{dep}} = 5\ \text{s}$.

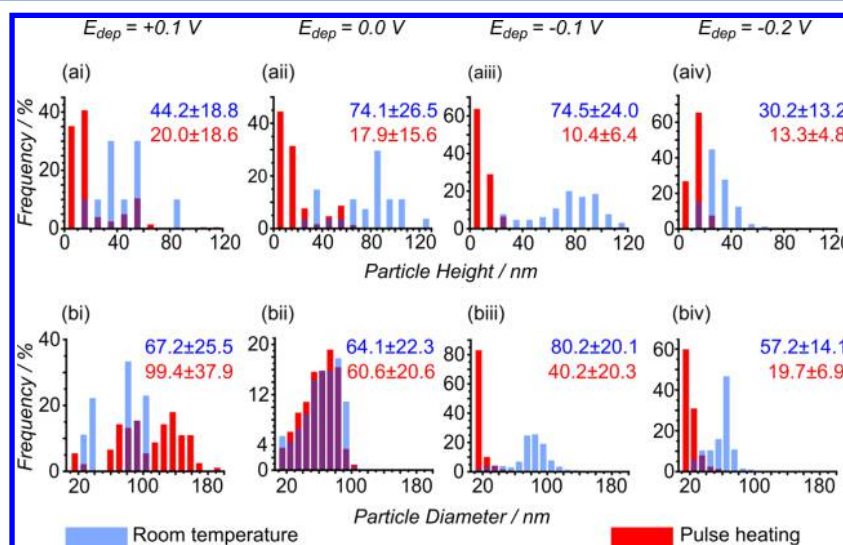


Figure 6. Histogram analysis of Pt NP (a) height and (b) diameter for both RT (blue) and laser pulse heating (red) experiments extracted from AFM and FE-SEM images, respectively. Electrodeposition was carried out for $t_{\text{dep}} = 5\ \text{s}$ at $E_{\text{dep}} = +0.1\ \text{V}$ (ai and bi), $0\ \text{V}$ (a-ii and b-ii), $-0.1\ \text{V}$ (a-iii and b-iii), and $-0.2\ \text{V}$ (a-iv and b-iv).

no evidence of elemental Pt dissolution or oxide layer formation is observed in these CVs.⁴⁸

Similar features are observed under laser pulse heating, as shown in Figure 3b, with the current registered for peaks 1–3 increasing with further cycling. The charge transferred during either H_{ads} or H_{des} (peaks 2 and 3, respectively) from the electrodeposited Pt provides an estimate of the electroactive surface area (ESA), using a value of $210\ \mu\text{C}\ \text{cm}^{-2}$ for monolayer hydrogen adsorption.⁵² Table S1 in the Supporting Information shows H_{des} and the associated ESAs for consecutive scans under both RT and increased temperature

conditions, quantitatively demonstrating the increased surface density of Pt in the presence of laser pulse heating.

3.2. Chronoamperometric Electrodeposition of Pt under Room-Temperature and Laser-Heated Conditions. To further investigate the Pt electrodeposition process, chronoamperometric current–time (i – t) plots were first recorded from a potential where no Faradaic electron transfer takes place ($+0.75\ \text{V}$) to different electrochemical driving forces ($E_{\text{dep}} = +0.1, 0, -0.1$, and $-0.2\ \text{V}$), within the Pt electrodeposition window (Figures 2 and 3) but before significant H_2 evolution. A fixed electrodeposition time ($t_{\text{dep}} = 5\ \text{s}$) was employed for both RT ($T = 20.0\ ^\circ\text{C}$) and laser pulse

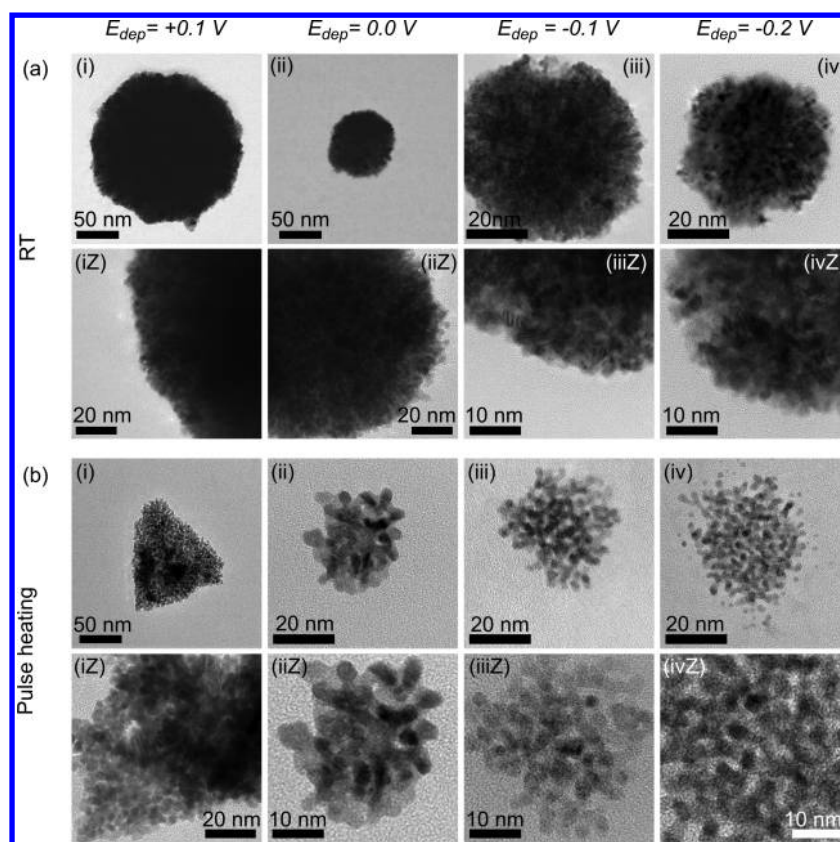


Figure 7. TEM images of Pt NPs obtained at (a) RT and (b) high temperature at E_{dep} values of (i) +0.1 V, (ii) 0 V, (iii) −0.1 V, and (iv) −0.2 V for $t_{\text{dep}} = 5$ s. (iZ–ivZ) Higher magnification TEM images of the Pt NPs.

heating ($T = 66.4$ °C, 10 ms on and 90 ms off), as shown in Figure 4.

Under RT conditions at the lowest driving potential (+0.1 V), after an initial decay in the charging current, the current appears almost Cottrellian in response, decreasing slightly with time. At $E_{\text{dep}} = 0$ V, after 1.5 s the current starts to rise with time, corresponding to the growth of Pt deposits which are initially sufficiently well separated to prevent individual diffusion fields from overlapping. This behavior is also evident in the −0.1 V data; however, it rises to a maximum and then decreases with time. This is due to the overlap of diffusion fields of neighboring particles, so that a predominantly linear diffusion field is established, resulting in the observed i – t characteristic. At the most negative potential, $E_{\text{dep}} = -0.2$ V, the current maximum is now occurring at a much shorter deposition time, due to the increased driving potential, with the i – t behavior approaching a Cottrellian-like response at longer times.⁵³

Under laser pulse heating, i increases with t regardless of the value of the electrodeposition potential and the currents are greater than those observed under RT conditions. Due to the pulsed nature of the heating procedure, it is difficult to interpret the i – t behavior in the same way as it is for RT. Therefore, microscopy analysis was employed to understand the nucleation and growth behavior in further detail under both temperature conditions. It is important to note that H_{ads} and H_2 evolution starts at slightly less negative potentials under high-temperature conditions (H_2 evolution onset −0.19 V) in comparison to RT (H_2 evolution onset −0.21 V) (Figure 2). This further suggests that, at the highest E_{dep} , the increase in

current could be related to both Pt deposition and simultaneous H_2 evolution.

3.3. Microscopic Analysis of Pt NPs Electrodeposited at Room and High Temperatures. Representative (i) AFM and (ii) FE-SEM images recorded for Pt electrodeposition on BDD under both (a–d) RT and (e–h) laser pulse heating conditions for a fixed t_{dep} value of 5 s and E_{dep} values of +0.1, 0, −0.1, and −0.2 V are shown in Figure 5. The images provide statistical data (from $n = 3$ images) on NP height and diameter extracted from AFM and FE-SEM images, respectively, as shown in the histogram analysis in Figure 6. FE-SEM was used to extract diameter information, as AFM can slightly overestimate the true diameter due to AFM tip convolution effects.⁵⁴

From Figures 5 and 6 it is immediately clear that irrespective of the temperature an increase in E_{dep} increases the particle number density; with an increase in temperature resulting in a higher number density in comparison to RT deposition, for all E_{dep} values considered. Number densities are given quantitatively in Figure S2 in the Supporting Information. At elevated temperatures, it is also clear that, as E_{dep} becomes more negative, the particle size distribution moves away from being heterogeneous toward more homogeneous, which is also significant given the polycrystalline nature of the BDD surface. The more elevated grains in the AFM images (corresponding to the lighter grains in the FE-SEM images) represent the lower doped regions and vice versa.⁵⁵ Previous studies have found a difference in the boron doping levels of ~ 4 , for similar material.⁵⁶ Under laser pulsed heating with the most negative E_{dep} (i.e., largest overpotential), the size of the Pt particles is the

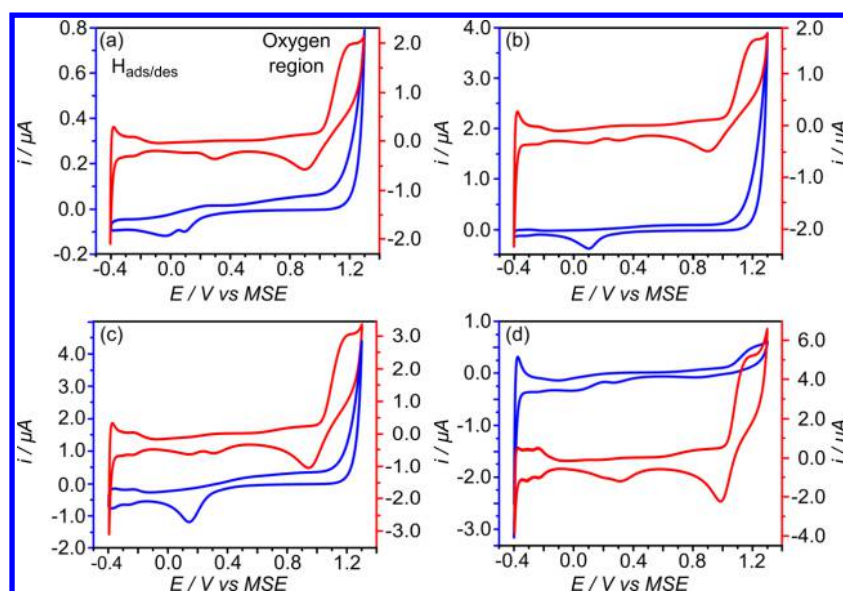


Figure 8. CVs of Pt NPs supported on BDD electrodes in deaerated 0.5 M H_2SO_4 solution at 0.05 V s^{-1} at RT. Electrodes were made by electrodeposition of Pt on BDD under (blue) RT and (red) laser pulse heating conditions using E_{dep} values of (a) +0.1 V, (b) 0 V, (c) −0.1 V, and (d) −0.2 V, respectively, at $t_{\text{dep}} = 5 \text{ s}$. Note that the y scales have been offset for clarity.

smallest of all, with the most homogeneous distribution (mean particle height of $13.3 \pm 4.8 \text{ nm}$).

RT electrodeposition of Pt, using a single potential step (SPS), often shows a wide distribution of particle sizes; such growth is often referred to classically as “progressive” nucleation.⁵⁷ More recently, the term aggregation has also been used to describe this process, thought to be due to the physical coalescence of growing clusters and the overlapping of diffusion fields of neighboring NPs. DPS chronoamperometry has proven effective as a means of controlling the monodispersity in the size distribution for Pt NPs on HOPG or glassy carbon electrodes.^{31,58–60} However, given the inherent surface heterogeneity in polycrystalline BDD, more extreme procedures were thought to be necessary to achieve monodispersity: for example, the use of a combination of wet chemical seeding of Pt NPs on the surface, followed by electrodeposition.^{61,62}

In contrast, our data show that, by implementing laser pulse heating during the electrodeposition process, SPS chronoamperometry, at the highest overpotential, is sufficient to significantly reduce the variation in particle size and overcome variations in dopant density in the underlying BDD. We attribute this, in part, to the large overpotential and the rapid heating/cooling of the electrode interface acting to separate the nucleation and growth processes sufficiently over the time scale of electrodeposition. Note that at this potential simultaneous H_{ads} and H_2 evolution is also occurring to the greatest degree, which could also play a role. For example, recently, Moffat et al. electrodeposited a single monolayer of Pt using these processes to quench subsequent growth.⁶³

3.4. Investigation of the Structure of Pt NPs Using TEM Imaging. Even though AFM and FE-SEM provide high-resolution topographical information, they do not have the resolving power to extract details on NP structure and crystallinity. Further insight was thus obtained by imaging the Pt NPs using TEM, as shown in Figure 7, where the particle morphology was investigated to examine the nature of atomic cluster aggregation/agglomeration and the crystallographic arrangement.

Figure 7a,b shows TEM images of typical Pt NPs obtained under both (a) RT and (b) laser pulse heating conditions for the four different electrodeposition driving forces. The elemental composition was confirmed by EDXS as shown in Figure S3 in the Supporting Information, suggesting that only Pt, not PtO_x , is present and the particles are free from any potential contamination during transfer.

Immediately apparent in Figure 7 is the difference in the structures formed under the two different temperature conditions. For both, the particles are clearly formed from much smaller nanoparticles, which themselves are composed of clusters of atoms that aggregate together. Under RT conditions, the Pt NPs appear to aggregate much more tightly in a more spherical or quasi-spherical structure. In contrast, under laser pulse heating conditions the Pt particles are less well packed and more porous in nature. TEM shows that they appear to be composed of aggregated Pt NPs, 2–5 nm in diameter: the more negative E_{dep} , the more loosely packed the structure. Similar morphologies have been obtained using solution-based synthesis approaches. Such structures are described in the literature as nanoporous or porous structures of high catalytic activity.^{31,33,34}

Selected area electron diffraction (SAED) shows that the Pt aggregates made by both procedures are polycrystalline (Figure S4 in the Supporting Information).⁶⁴ Under RT conditions, it is possible to assign the visible (hkl) indices to (111), (210), (310), (321), (200), (300), (320), (220), (311), and (331) of fcc Pt. Under elevated temperature conditions, the following (hkl) indices are observed: (111), (200), (220), (222), (422), (333), (511), (310), (330), (311), (322), (320), and (410). Thus, the particles formed under higher temperature conditions have more of the higher index facets. These are thought to have a larger effect on increasing the catalytic activity, in comparison to lower index planes, due to the greater presence of dangling atomic bonds, steps, edges, and kink sites.^{65,66}

From the AFM and FE-SEM data (Figures 6 and 7), the main effect of pulsing the temperature, in combination with increasing E_{dep} , was shown to be decreasing particle size and increasing monodispersity during electrodeposition. Crucially,

TEM now reveals pulsing the temperature plays a key role in the creation of higher index facets during electrodeposition and increasing the porosity of the resulting structure. Interestingly, the occurrence of most of the higher index facets occurs under laser pulsed conditions at the lowest E_{dep} (Figure S4 in the Supporting Information), while qualitatively, the porosity appears greatest at the highest (most negative) E_{dep} . The latter could also have a contribution from depositing at a potential into the H_{ads}/H_2 evolution potential region. At these higher negative potentials and in the presence of high temperature, a decrease in the solution viscosity and surface tension results,⁶⁷ allowing H_2 bubbles to break away from the surface with smaller diameters, potentially facilitating transport of these detached H_2 bubbles through the interior of the deposit.⁶⁸

3.5. Electrocatalytic Performance and Methanol Oxidation Studies. The electrochemical response of the electrodes produced under both temperatures using all four electrodeposition driving forces was investigated in deaerated 0.5 M H_2SO_4 . In electrocatalysis, size, structure, and crystallographic orientation are important.⁶⁹ Size influences the flux of reactant/product toward/away from the surface, while structure and crystallographic orientation can influence the binding ability of counterions in solution and reactants/intermediates or products during inner-sphere (electrocatalytic) electron transfer.^{70,71} For example for MeOH oxidation in acidic media the electroactivity of Pt, low-index, single-crystal planes was found to be (100) (greatest) > (110) > (111) (lowest).⁷²

When CVs are recorded in acidic solutions, the $H_{\text{ads}}/H_{\text{des}}$ surface oxidation, and oxygen adsorption/desorption ($O_{\text{ads/des}}$) features can also be used to highlight differences in the electroactivity of the electrodeposited aggregates. These responses are shown in Figure 8, where CVs were recorded by cycling the Pt NP-BDD electrodes in 0.5 M H_2SO_4 at a scan rate of 0.05 V s^{-1} versus MSE until a consistent response was obtained.

The characteristic voltammetric features associated with Pt are visible in the acidic electrolyte, as discussed previously for Figure 3. The only difference here (Figure 8) is that the Pt NPs are preformed on the electrode surface, removed from the electrodeposition solution, and then cycled in acid. In general, the CVs confirm the polycrystalline nature of the electrodeposited Pt. Additionally, the shape and the charge passed due to $H_{\text{ads/des}}$ depend upon the size and structure of Pt NPs. Pt particles made by applying low driving forces (+0.1 V) at RT show one $H_{\text{ads/des}}$ peak. In contrast, at RT, Pt particles produced under high driving force conditions (0.0, −0.1, and −0.2 V) show two peaks in the $H_{\text{ads/des}}$ regions. Under laser pulse heating electrodeposition different CV signatures are observed in comparison to RT. Two $H_{\text{ads/des}}$ peaks are observed for $E_{\text{dep}} = +0.1$, 0, and −0.1 V, while three peaks in the H_{des} region and two peaks in the H_{ads} regions are visible at $E_{\text{dep}} = -0.2$ V.

The $H_{\text{ads/des}}$ peaks are usually associated with different forms of adsorbed H and the contribution of different crystal facets,^{31,73} with the position on the voltage axis and number of peaks correlating with the dominant crystal planes present. Using single-crystal electrodes, for the low-index crystal planes, it has been found that the $H_{\text{ads/des}}$ peaks occur in the order (110) > (111) > (100),⁷³ with (110) being the most negative on the potential axis due to the weakly bonded state of H, whereas (100) promotes the strongest binding of H.⁷⁴ Labeling of the $H_{\text{ads/des}}$ features in Figure 8 to specific crystal planes is challenging, especially given the SAED data in section S15 in

the Supporting Information, which illustrate the wide variety of different crystallographic orientations initially present prior to cycling in acid.

All Pt NPs electrodeposited under laser pulse heating conditions showed a greater ESA (from the H_{des} data) in comparison to RT electrodeposition conditions, as shown in Table 1. This could be due to a combination of factors; for

Table 1. Experimental Values of the Charge Associated with H_{des} on Pt NPs and Measurement of the Corresponding ESA

E_{dep}/V	temp/ $^{\circ}\text{C}$	$Q_{H_{\text{des}}}/\mu\text{C}$	ESA/ cm^{-2}
+0.1	20.0	0.00183	8.69×10^{-6}
	66.4	0.02102	1.00×10^{-4}
0.0	20.0	0.00253	1.21×10^{-5}
	66.4	0.03658	1.74×10^{-4}
−0.1	20.0	0.06178	9.93×10^{-5}
	66.4	0.05636	2.68×10^{-4}
−0.2	20.0	0.02904	1.38×10^{-4}
	66.4	0.12170	5.80×10^{-4}

example, aggregation into a more compact structure, as seen for RT electrodeposition, leads to a decrease in the ESA. Furthermore, laser heating always leads to an increase in the NP number density (for the same E_{dep}) (Figure S2 in the Supporting Information).

In the oxygen adsorption/desorption region, $O_{\text{ads/des}}$ (+0.7 to +1.3 V), the currents passed are also very different. There are significant $O_{\text{ads/des}}$ currents associated with the laser pulsed deposited NPs at all E_{dep} values. In contrast, only NPs produced at RT, at the most negative E_{dep} , show an $O_{\text{ads/des}}$ current. This suggests that laser heating, in all cases, results in electrochemically accessible atomic arrangements, such as the higher index facets observed in SAED, which are more efficient at promoting oxygen adsorption in this potential region.^{65,75}

The electrochemical catalytic activity of the densely packed “compact” and “nanoporous” Pt particles for the oxidation of MeOH, i.e. the methanol oxidation reaction (MOR), was also investigated using CV analysis. The ideal MOR catalyst is one which is able to efficiently oxidize MeOH, decompose water (to produce adsorbed OH which acts as a further oxidizing agent), and weakly bind poisoning species, such as CO,⁷⁶ which inhibit further MeOH oxidation. CVs were measured in a deaerated aqueous solution of MeOH (CH_3OH , 1 M) and sulfuric acid (H_2SO_4 , 0.5 M), between 0 and +1 V, as shown in Figure 9.

All MOR CVs show similar features where there is a peak current response due to MeOH oxidation on the forward scan (starting at 0 V vs SCE and ending at +1.0 V vs SCE)⁷⁷ and a prominent oxidative peak in the backward scan. The latter has been traditionally associated with residual (intermediate) carbon species poisoning the surface^{78,79} (in particular CO_{ads}) rather than to the oxidation of freshly chemisorbed species. As such, the ratio of the forward peak current, $i_{\text{p,f}}$, to the back peak current, $i_{\text{p,b}}$, has been used extensively in the literature as a guide to the ability of the catalyst to tolerate CO poisoning. The more residual adsorbed species on the surface, the more the catalyst is poisoned and the lower the $i_{\text{p,f}}/i_{\text{p,b}}$ ratio. Very recent preliminary research, however, has suggested that the back current does not reflect CO adsorption and poisoning⁸⁰ but instead is indicative of Pt surface oxygenation and degree of oxophilicity.⁸¹

In order to compare the difference in the electrocatalytic activity of the different Pt NPs toward MeOH oxidation, the

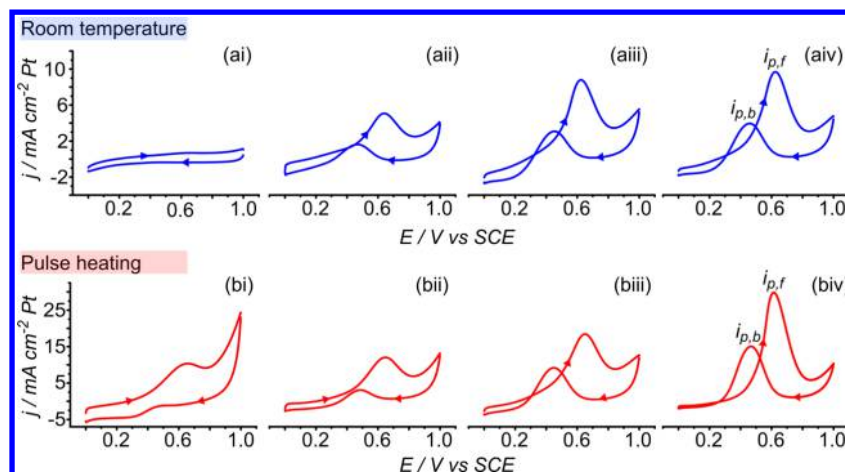


Figure 9. CVs recorded at 0.05 V s^{-1} for the oxidation of 1 M MeOH in sulfuric acid ($0.5 \text{ M H}_2\text{SO}_4$), catalyzed by Pt NPs-BDD electrodeposited under (a) RT and (b) laser pulse heating conditions for E_{dep} values of (i) $+0.1 \text{ V}$, (ii) 0 V , (iii) -0.1 V , and (iv) -0.2 V and $t_{\text{dep}} = 5 \text{ s}$.

current passed is normalized with respect to the Pt NP surface area (ESA, calculated from H_{des} , displayed in Table 1) and then evaluated with respect to both $i_{p,f}$ and $i_{p,b}$ and the ratio of $i_{p,f}$ to $i_{p,b}$ (Table 2).^{82–84} Higher $i_{p,f}$ corresponds to a greater catalytic activity,⁶⁵ and if we take the traditional view, a higher ratio indicates a catalyst less subject to CO poisoning.⁸⁵

Table 2. Experimental Measurement of the Values of the Current Corresponding to MeOH Oxidation and the Tolerance of Pt NP Catalyst toward CO Adsorption

E_{dep}/V	temp/ $^{\circ}\text{C}$	$i_{p,f}/\text{mA cm}^{-2}$	$i_{p,b}/\text{mA cm}^{-2}$	$i_{p,f}/i_{p,b}$
+0.1	20.0	0.68	0.36	1.8
	66.4	10.36	1.18	8.7
0.0	20.0	5.08	1.66	3.1
	66.4	12.02	3.07	3.9
−0.1	20.0	8.79	3.08	2.8
	66.4	17.61	8.42	2.1
−0.2	20.0	9.68	3.95	2.4
	66.4	29.84	15.04	1.9

Noticeably, both the forward ($i_{p,f}$) and the backward ($i_{p,b}$) peak current densities for the MOR increase with an increase in the electrodeposition driving force for Pt NP-BDD electrodes prepared under both RT and laser pulse heating conditions. Previous reports have attributed enhancements in $i_{p,f}$ to the presence of specific crystallographic orientations, defects in the crystalline structure, and the structure of the intergrain boundaries.^{43,72,86} All of the more “porous” Pt particles, from pulse laser electrodeposition, show higher $i_{p,f}$ values in comparison to those formed under RT, which we attribute to both a greater accessibility to, and increased number of, catalytically active sites,⁸⁷ as also evidenced by the TEM and SAED data (Figure 7 in the main text and Figure S4 in the Supporting Information) and the CV data in Figure 8. The largest $i_{p,f}$ value is found for the nanoporous Pt particles electrodeposited under laser pulse heating conditions at the highest overpotential (-0.2 V).

The highest $i_{p,f}/i_{p,b}$ value (Table 2) is found for Pt NPs synthesized at the lowest driving potential ($+0.1 \text{ V}$) under laser pulse heating conditions ($i_{p,f}/i_{p,b} = 8.7$). Although these particles show the most higher index planes (from SAED), they are less porous (from TEM) in comparison to those particles produced at more negative E_{dep} values under laser pulsed conditions.

Thus, whether all the identified crystal planes can be accessed is not known, due to the lower porosity. Note at this deposition potential the $i_{p,f}$ value for the MOR is still higher than all values recorded on Pt particles deposited under RT conditions. Although catalytic, electrochemically accessible surfaces are preferred,⁸⁸ to produce the maximum $i_{p,f}$ such surfaces may also promote very efficient adsorption of residual MeOH oxidation intermediate products which can cause a poisoning effect. This may explain the data associated with the Pt NPs deposited under laser pulsed heating conditions at the largest overpotential. These particles show the highest $i_{p,f}$ data but also return the lowest $i_{p,f}/i_{p,b}$ ratio ($i_{p,f}/i_{p,b} = 1.9$) (Table 2), if the recognized view is upheld.

4. CONCLUSIONS

A new approach to producing nanoporous Pt NPs of high electrocatalytic activity has been introduced: namely, electrodeposition in tandem with continuous heating and cooling of the electrode surface. The latter is achieved using a pulsed laser in conjunction with a high thermal conductivity electrode: boron-doped diamond. Injecting bursts of heat via IR laser modulation into the interface and then rapidly cooling acts to separate the nucleation and growth processes, leading to NP size homogeneity and the production of “porous”, electrochemically accessible structures. TEM and SAED showed the particles to be composed of loosely packed Pt aggregates, 2–5 nm in size, containing high-index crystal planes. In contrast, electrodeposition at RT resulted in much more densely packed “compact” structures.

Investigation of NP activity toward MeOH oxidation revealed a significantly higher catalytic activity for the nanoporous NPs prepared under laser heated conditions, in comparison to those electrodeposited at RT. The highest activity was found for the Pt NPs electrodeposited under laser heated conditions at the highest driving potential. Qualitatively, TEM also revealed these NPs to be the most porous. However, the ratio of $i_{p,f}$ to $i_{p,b}$ was highest for NPs deposited under laser heated conditions but with the smallest overpotential. If we assume the highest ratio indicates the NPs least susceptible to adsorbed CO poisoning, the data suggest that the most catalytically active NPs may also encourage binding of residual adsorbed CO and that a compromise must be reached.

Future work will look to further explore the role of temperature pulsing during electrodeposition on the resulting structure and catalytic activity of NPs, given that temperatures in excess of 100 °C in aqueous solutions are possible²⁵ using this approach.

■ ASSOCIATED CONTENT

■ Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acscatal.7b02701.

Description of the replication procedure employed to transfer Pt NPs from the BDD electrode to a Cu TEM grid. Experimental values of the charge passed during H_{des} and the associated ESAs for the consecutive CV scans of Pt electrodeposition under RT and laser pulse heating conditions. Statistical AFM analysis of Pt NP density, electrodeposited under RT and laser pulse heating conditions. EDXS elemental analysis of Pt NPs transferred to TEM grids. Selected area electron diffraction (SAED) patterns of Pt NPs electrodeposited under RT and laser pulse heating conditions (PDF)

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Notes

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